COMP6713 Group Project

Evaluation and Analysis Report

Team name: Never Fail and I Love NLP

1. Project Scope

The table below shows the project scope of our group.

PART A: Problem Definition **Minimum: 10 credits**			**Fill your project details here**
	Points	**Details**	
NLP Problem	5 credits per problem	Examples: Question-answering, Sentiment Analysis, etc. (Refer to course modules)	Sentiment Analysis: binary (positive/negative) classification of movie reviews.
Text Source/Domain	5 credits per text source/domains	Examples: News Articles, Medical papers, social media posts, etc.	Social media (Reddit) + movie reviews (IMDb). Two distinct domains for cross-domain evaluation.
PART B: Dataset Selection **Minimum: 20 credits**			
Use two existing datasets		Publicly available datasets.	1. GoEmotions (Demszky et al., ACL 2020): ACL Anthology | HuggingFace Reddit comments, 27 emotion labels → collapsed to pos/neg for binary task. Used as training set. 2. IMDb Large Movie Review (Maas et al., ACL 2011): ACL Anthology | HuggingFace Out-of-domain test set for cross-domain evaluation. Confirmed acceptable by supervisor.
Create your own labelled dataset	20 credits	Correct selection of labels, inter-annotator agreement	None
Use an existing lexicon	10 credits	Examples: WordNet, medical ontology	SentiWordNet (via NLTK) lexicon nltk.org
PART C: Modelling			
Implement a rule-based or statistical model as a baseline	-	Essential. Please keep this approach simple – this is only a baseline.	SentiWordNet lexicon lookup: sum positive/negative scores per token, threshold to classify.
Minimum: 30 credits			
Use an existing pre-trained/fine-tuned model	5 credits per model	You must compare the performance of multiple models to accrue credits, if you are only using	Model A — cardiffnlp/twitter-roberta-base-sentiment Model B — distilbert-base-uncased-finetuned-sst-2-english

		available fine-tuned models or only prompting.	
Fine-tune a model based on a dataset	20 credits per out-of-the-box fine-tuning method	Examples: fine-tuned BERT, prefix-tuned LLaMA	1. Fine-tune DistilBERT on GoEmotions (binary labels). Colab free-tier sufficient. Reference: TweetNLP 2. Fine-tune RoBERTa on GoEmotions(binary labels). Trained on RTX 4090.
Extend a method	30 credits per extended fine-tuning method	Examples of extension: modification of the loss function, incorporation of structured ontology, prompting method other than zero/few-shot prompting, etc.	None
Integrate a language model with external tools	20 credits	Usage of a library like LangChain/LangSmith	Weights & Biases (W&B) for hyperparameter search: wandb.ai Sweep over learning rate, batch size, epochs. Log and compare runs via W&B dashboard.
	PART D: Evaluation **Minimum: 20 credits**		
Quantitative Evaluation	10 credits	Appropriate metrics.	Accuracy, F1, Precision, Recall. Cross-domain evaluation: train on GoEmotions → test on IMDb (out-of-domain). Domain shift discussed in report.
Qualitative Evaluation	5 credits	Examine mis-classified instances and produce common error types	Manually inspect 20–40 errors; categorise: negation, sarcasm, domain mismatch.
Command line testing	5 credits	Interface to test out the system. This can be executed using an input argument or input file.	Python CLI (predict.py --text "..." or --file reviews.txt).
Demo	10 credits	A simple demo through Gradio (or equivalent).	Gradio app: text input → sentiment label + confidence; model selector (baseline / pre-trained / fine-tuned).

2. Introduction

This report evaluates a binary sentiment analysis system that classifies text as either positive or negative. The project uses GoEmotions that is a Reddit-based dataset, as the source domain for training, and the IMDb movie review dataset as the target domain for testing. This cross-domain

setup is intentional: it tests whether models trained on short social media comments can generalise to longer, more complex movie reviews.

This project compares five sentiment analysis models: a SentiWordNet baseline as the rule-based reference, two zero-shot models (RoBERTa and DistilBERT-SST2) and two models fine-tuned on the GoEmotions binary dataset, namely DistilBERT and RoBERTa.

The project also includes a Gradio web demo (demo/app.py) and a command-line inference tool (demo/predict.py) for interactive testing. The qualitative error analysis examines 30 misclassified examples from each of the two fine-tuned models(DistilBert and RoBERTa).

3. Experiment Setup

3.1 Models

3.1.1 SentiWordNet Baseline (models/baseline.py).

This model assigns a sentiment label by summing positive and negative word scores from the SentiWordNet lexicon. It uses NLTK for tokenisation and POS tagging. Because it only looks at individual words, it has no understanding of context, negation, or irony.

3.1.2 RoBERTa (zero-shot) (models/pretrained_roberta.py).

This is the cardiffnlp/twitter-roberta-base-sentiment model, originally trained on Twitter data. It is applied to the IMDb test set without any additional fine-tuning. Twitter and movie review text differ significantly in length and style.

3.1.3 DistilBERT-SST2 (zero-shot) (models/pretrained_distilbert.py).

This is the distilbert-base-uncased-finetuned-sst-2-english model, which was originally trained on SST-2 — a movie review sentiment dataset. It is also applied without further fine-tuning. Because SST-2 is in the same domain as IMDb, this model has a natural advantage over the others.

3.1.4 DistilBERT (fine-tuned) (models/finetune_distilbert.py).

A DistilBERT model (distilbert-base-uncased) fine-tuned on the GoEmotions binary training set using the HuggingFace Trainer API. Hyperparameter search was conducted via a Weights & Biases grid sweep over learning rate {2e-5, 3e-5} and number of epochs {2, 3}, totalling 4 runs. The best configuration — learning rate 3e-5, 2 epochs — achieved F1 = 0.929 and accuracy = 0.908 on the GoEmotions validation set. The model was trained on Google Colab (T4 GPU) and the checkpoint is saved in models/distilbert-finetuned/best_model/. It is evaluated on IMDb as an out-of-domain test set.

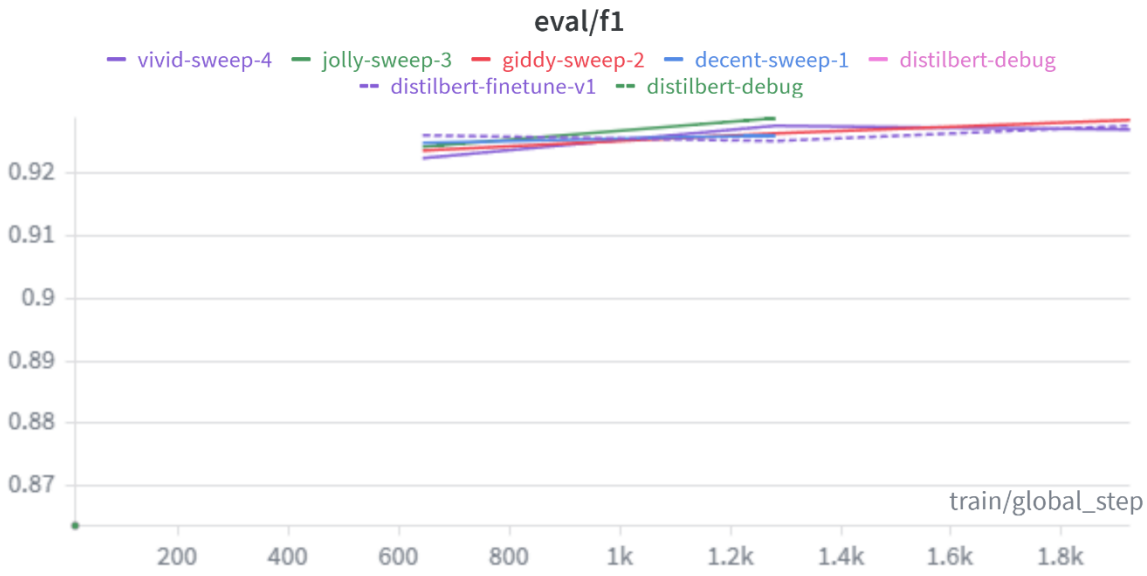

Figure 1: W&B sweep results showing eval/F1 across 4 hyperparameter configurations.

3.1.5 RoBERTa (fine-tuned) (models/finetune_roberta.ipynb).

We trained a RoBERTa model (roberta-base) by utilizing the GoEmotiions binary dataset by fine-tuning the model, while the HuggingFace Trainer API supports the Transformers framework. In addition, a hyperparameter search was performed, which used Weights & Biases to implement Bayesian search over learning rate (log-uniform in [1e-5, 5e-5]), batch size {32, 64, 128}, number of epochs {2, 3, 4}, weight decay {0.0, 0.01, 0.1}, and warmup ratio (uniform in [0.0, 0.2]), totalling 20 runs.

The optimal configuration was learning rate = 4.08e-05, batch size = 64, epochs = 4, weight decay = 0.1, and warmup ratio = 0.079. It achieved F1 = 0.939 and accuracy = 0.920 in the GoEmotions validation set. The model was trained on an RTX 4090 (24 GB) and the checkpoint is saved in models/Roberta-finetuned/best_model/. It is evaluated on IMDb as an out-of-domain test set.

Figure 2: W&B sweep results showing eval/F1 across 4 hyperparameter configurations.

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3.2 Tools and Environment

The project was developed in Python 3.10/3.11 and uses the following tools:

- Hugging Face Transformers — loading and running all pre-trained and fine-tuned models
- scikit-learn — computing accuracy, precision, recall, F1-score, and confusion matrices
- NLTK + SentiWordNet — lexicon lookup and POS tagging for the baseline
- Weights & Biases (W&B) — hyperparameter sweep for both fine-tuned models
- Gradio — interactive web demo (demo/app.py)
- Git LFS — storing and distributing fine-tuned model checkpoints

3.3 Dataset

The evaluation uses the IMDb test set with 25,000 reviews - 12,500 positive and 12,500 negative. Both datasets are downloaded automatically by running data/build_dataset.py. The GoEmotions training data has 28 emotion labels, which are mapped to binary sentiment (positive / negative) using data/label_mapping.py before fine-tuning.

This is a cross-domain evaluation:

- Source domain(training): GoEmotions — short Reddit comments, mapped to binary sentiment
- Target domain(testing): IMDb — long-form movie reviews with binary labels

3.4 Evaluation Metrics

All five models were evaluated using accuracy, precision, recall, and F1-score. Precision, recall, and F1 are reported for the positive class. Batch predictions were generated by

running models/run_predictions.py on the IMDb test set and stored in JSON files for comparison.

4. Empirical Evaluation

4.1 Results

Table 1 shows the performance of all five models on the IMDb test set.

Model	Accuracy	Precision	Recall	F1
SentiWordNet Baseline	0.6107	0.5704	0.8964	0.6972
RoBERTa (zero-shot)	0.8223	0.9242	0.7022	0.7981
DistilBERT-SST2 (zero-shot)	0.8921	0.9151	0.8644	0.8890
DistilBERT (fine-tuned)	0.7523	0.7197	0.8266	0.7695
RoBERTa (fine-tuned)	0.7744	0.7220	0.8925	0.7983

Table 1Performance on the IMDb test set Precision, Recall, and F1 for the positive class.

4.2 Discussion

The best performing model in this project is DistilBERT-SST2 (zero-shot). Its accuracy is 89.2% and its F1 score is 0.889. The main reason for this is domain alignment. SST-2 that was used in this project is made up of movie reviews. This is the same domain as IMDb. So the model already understood how sentiment is expressed in movie reviews without any fine-tuning in this project. This result suggests that picking the right pre-training domain can matter more than fine-tuning, especially in a cross-domain setting.

The weakest one is SentiWordNet Baseline. At 61.1% accuracy, the baseline performs the worst. It has very high recall (0.896) but low precision (0.570), which means it over-predicts positive too often. This is a known weakness of lexicon-based approaches — they have no way to handle context. When a reviewer writes "not a great film," the model only sees "great" and labels it positive. In long, complex movie reviews where mixed language is common, this approach breaks down quickly.

DistilBERT-SST2 clearly outperformed RoBERTa zero-shot (89.2% vs 82.2%). RoBERTa was pre-trained on Twitter data, which is very different from movie reviews in both length and style. It also shows a skewed pattern, high precision (0.924) but low recall (0.702), meaning it often fails to recognise positive reviews, being too conservative with positive predictions.

Both fine-tuned models performed worse than DistilBERT-SST2 zero-shot. Fine-tuned DistilBERT reached 75.2% accuracy and fine-tuned RoBERTa reached 77.4%. While fine-tuning on GoEmotions improved over the SentiWordNet baseline, it did not help

enough to beat the in-domain zero-shot model. Fine-tuning appears to have pushed both models toward the patterns of short Reddit comments, which do not transfer well to the longer, more structured writing of IMDb reviews. It is also worth noting that fine-tuned RoBERTa performed slightly better than fine-tuned DistilBERT. This could be because RoBERTa is a larger model with stronger pre-training. Even after fine-tuning on GoEmotions, those advantages may have carried over.

5. Cross-Domain Evaluation Discussion

An important finding can be found from the result is that fine-tuning on GoEmotions is not a very good way to improve IMDb performance. This situation is worth studying more carefully. It points to a key challenge in cross-domain NLP.

GoEmotions reviews are one or two short sentences from Reddit. IMDb reviews are longer and their comments are more complex and includes several references from movies, sometimes mixing positive and negative language before giving their main idea of a moive. A model trained on short text learns to classify from a few words, which fails when the signal is spread across hundreds of words.

GoEmotions has 28 fine-grained emotion categories that had to be collapsed into binary positive/negative labels using data/label_mapping.py. For example, surprise, curiosity, or confusion do not map cleanly to either class. This is noise into the training data that can weaken the fine-tuned model to make reliable predictions.

In terms of writing style and vocabulary, the comments in Reddit are informal and direct in the writing. The reviews in IMDb are in a very wide range in the style from casual fan writing to detailed critical review. Some reviewers talk about the history of a film, compare it with others or write in a sarcastic or understated style. These patterns are not present at all in the GoEmotions training data.

Many IMDb reviewers express their opinion indirectly through negation, irony, comparisons to other films, or plot descriptions that contain no explicit sentiment words. These patterns require discourse-level understanding, which models trained on short social media text are not well prepared for.

Taken together, these factors explain why DistilBERT-SST2, a model that was never fine-tuned in this pipeline, outperformed both fine-tuned models by a substantial margin. The domain gap between GoEmotions and IMDb is too large for standard fine-tuning to fully bridge.

6. Qualitative Evaluation

6.1 Setup

To understand where the fine-tuned models fail, we manually examined 30 misclassified examples from each fine-tuned model (30 from fine-tuned DistilBERT and 30 from fine-tuned RoBERTa) giving 60 samples in total. Each example was assigned to one of five error categories:

- Negation: the model failed to follow a negation or a 'but' that reversed the overall sentiment
- Sarcasm: ironic language tricked the model into a wrong prediction
- Domain mismatch: the writing style, genre, or topic fell outside typical film review conventions
- Ambiguous: the sentiment is genuinely unclear, even to a human reader
- Other: compound or unusual failures that do not fit the categories above

6.2 Error Distribution

Table 2 Shows the breakdown of errors for each model and the combined totals across all 60 samples.

Error Category	DistilBERT (n = 30)	RoBERTa (n = 30)	Combined (n = 60)	Combined%
Negation	3	4	7	12%
Sarcasm	6	4	10	17%
Domain Mismatch	4	7	11	18%
Ambiguous	11	8	19	32%
Other	6	7	13	22%
Total	30	30	60	100%

Table 2 Distribution of error categories across 60 manually analysed misclassified samples

Ambiguous cases were by far the most common error type overall (32%), which suggests that a large share of the models' mistakes come from reviews that are genuinely difficult to classify even for human readers. Domain mismatch (18%) and Other (22%) were the next most frequent, followed by sarcasm (17%) and negation (12%).

There are also some differences between the two models. Out of 30 cases, there are 6 cases of sarcasm error and 11 cases of ambiguous error in the Fine-tuned DistilBERT. Fine-tuned RoBERTa recorded 7 cases of domain mismatch, and 7 instances of Other errors. This difference due to architecture and vocabulary. Because one model focuses more on individual words while the other pays more attention to the broader context, so the difference between them is because of their architecture and vocabulary.

6.3 Error Category Analysis

6.3.1 Ambiguous (19 cases, 32%)

This type was the largest error category in this evaluation. These cases are more difficult because the reviews mix positive and negative language together. It is hard to identify comments' sentiment. In some cases, a human reader might misunderstand the original meaning of reviewers.

Example (error_analysis_distilbert.json sixth sentence) — DistilBERT(True: 0, Predicted: 1, Confidence: 0.5654)

Review excerpt: "This film is so bad it's hilarious... It is very amusing though, and I guess either way it could go down as a cult classic, either from being an amusing parody of the

Tarantino/Rodriguez style, or something that is very very unintentionally funny. Has to be seen to be believed."

This is a classic 'so bad it's good' review. The reviewer clearly finds the film entertaining despite its failures. The human label is negative, but the reviewer calls it amusing and potentially a cult classic, a positive prediction is arguably just as defensible. The low confidence (0.5654) shows the model itself was uncertain.

Example (error_analysis_roberta.json eighth sentence) — RoBERTa (True: 0, Predicted: 1, Confidence: 0.9976)

Review excerpt: "Maureen O'Hara is always beautiful and appealing... William Bendix is good, as is most of the supporting cast... It's a sweet movie but it doesn't feel sincere. I was hoping to be moved. I was, by O'Hara's gentle performance. But I don't like to feel manipulated."

Individual performances are appreciated, but the overall review is negative. RoBERTa focused on the appreciation for O'Hara and Bendix but overlooked the overall conclusion. The overconfidence (0.9976) indicates that RoBERTa concentrated too heavily on the positive vocabulary and failed to grasp the significant negative sentiment at the end.

6.3.2 Other (13 cases, 22%)

These mistakes display similar structures. The reviewer is often positive about something unrelated to the film being reviewed. It could be another remarkable piece of art, an actor's impressive filmography, or a general genre. But they are constructive about the specific subject. The model picks up the surrounding positive comments.

Example (error_analysis_distilbert.json second sentence) — DistilBERT(True: 0, Predicted: 1, Confidence: 0.9812)

Review excerpt: "An ear-splitting movie, a quasi-old-fashioned screwball romp... After an energetic animated credits sequence — which is much more fun than the rest of the picture — we have nothing to look at... the poor-choice supporting cast struggles to get laughs with lousy dialogue. It's an unfortunate set-back to the talents of director James Foley... Wretched. * from ****"

The review ends with a 1 out of 4 rating and the word 'Wretched,' yet the model predicted positive with near-maximum confidence (0.9812). The brief positive mention of the animated credits and the energetic opening description appear to have overwhelmed the clear negative verdict.

Example (error_analysis_roberta.json other part fifth sentence) — RoBERTa (True: 0, Predicted: 1, Confidence: 0.9856)

Review excerpt: "This is one of four 1970s movies by TV writer Lane Slate about sensationalistic murders in small towns. They feature likable TV personalities as police chiefs and quirky characters as town regulars, including light-touch love interests. The others are: They Only Kill Their Masters (James Garner, 1972)... On that level, they are

somewhat entertaining... But they suffer from overall thin stories and dialogue, slack pacing, bland settings, flat or exaggerated characters..."

The review is negative about this specific film, but the reviewer spends a large portion praising the other films in the same Lane Slate series, describing them as entertaining and making favourable cast comparisons. RoBERTa latched onto those positive comments about the similar films and predicted positive with high confidence (0.9776), missing that the reviewer's verdict on the actual reviewed film is clearly negative.

6.3.3 Domain Mismatch (11 cases, 18%)

Domain Mismatch errors happen when the review is written in a style, genre, or register that falls far outside what the model expects from a standard film or TV review. These are particularly common in the RoBERTa errors (7 out of 11 cases).

Example (error_analysis_distilbert.json domain_mismatch part first sentence) — DistilBERT (True: 0, Predicted: 1, Confidence: 0.9812)

Review excerpt: "I would like to comment on how the girls are chosen. why is that their are always more white women chosen then their are black women... you are just like the rest trying to act like you are so fair and nice. you are just a big fony hypocrite."

The text is not a review. This is a direct complaint about racial casting. A model trained on film quality reviews has no way to handle this kind of writing. Moreover, the words ‘fair’ and ‘nice’ pop up as accusations, but the model read them as positive signals. This caused the prediction to be positive. The confidence rate of 0.99 shows there was little to no language the model would classify as a typical negative review.

Example (error_analysis_roberta.json domain_mismatch part fifth sentence) — RoBERTa (True: 1, Predicted: 0, Confidence: 0.9912)

Review excerpt: "This movie gave me recurring nightmares, with Alan Rickman's voice representing an omnipotent, insidious, fascist ruler. The scariest movie I have ever seen — psychological terror more powerful than anything any 'horror' movie has ever achieved... This film is disturbing, but in a way that everyone should watch it."

The reviewer is praising this film precisely because of how disturbing and powerful it is. But RoBERTa read the text as negative, because words like 'nightmares,' 'terror,' 'fascist,' and 'disturbing' almost always appear in negative contexts. This is a case where the review genre — a serious political film critique — uses negative-sounding language as a form of praise, which the model was not trained to handle.

6.3.4 Sarcasm (10 cases, 17%)

Sarcasm errors happen when the model takes ironic language at face value. Both models made sarcasm errors, though fine-tuned DistilBERT made slightly more (6 vs 4 cases). The most common pattern is when a negative review contains one or two positive-sounding phrases that the model picks up on, while missing the ironic intent.

Example (error_analysis_distilbert.json sarcasm part first sentence) — DistilBERT (True: 0, Predicted: 1, Confidence: 0.5755)

Review excerpt: "This is a piece of Hollywood product that should have never left a film can. Dialogue without a plausible thought, plot without a point, staging without skill, directing without direction, and acting without the worth of some backwater high school's freshman class play. The entire cast should have been arrested for over acting. But otherwise, okay!"

The review goes through every part of the film and criticises all of them. Then it ends with "But otherwise, okay!" This closing line sounds positive, but it is clearly sarcastic. The model took it at face value and predicted positive. The confidence score was only 0.5755. This shows the model did notice the negative vocabulary in the rest of the review. However, that one sarcastic phrase at the end was still enough to change the final prediction.

Example (error_analysis_roberta.json sarcasm part first sentence) — RoBERTa (True: 0, Predicted: 1, Confidence: 0.5926)

Review excerpt: "'The Best Movie of the 90's' 'The Welsh Trainspotting'....Aye, right! I went into this movie with pretty high expectations, and it was all downhill from there... It is painfully bad at times."

The review does not consist of independent wording from the reviewer. Initially, two phrases, "Best Movie of the 90's" and "The Welsh Trainspotting" are quoted. Although the phrases appear to be positive, the reviewer then inserts their sarcastic opinion of the quotes with "Aye, right!" and the quotes are therefore being mocked. RoBERTa attributed the quoted phrases to be genuine positive opinions and therefore missed the mocking. The overall score for the quoted phrases was 0.5926. Therefore, the review's negative and mocking language provided some uncertainty for the model, although not enough to provide a different outcome.

6.3.5 Negation (7 cases, 12%)

Negation errors were the most infrequent kind across both models. DistilBERT recorded 3 errors of this result, and RoBERTa recorded 4. This style of error generally occurs when the model scrutinizes the words one at a time and loses big picture understanding. This can occur in some positive reviews, which may express a negative sentiment despite containing many negative-nuanced words about the subject. Alternatively, the model may overlook words of negation. This can occur due to the presence of negation words, such as 'not', 'didn't', and 'no', or the lack of these. In both situations, the model predicts the incorrect label.

Example (error_analysis_distilbert.json negation part third sentence) — DistilBERT (True: 1, Predicted: 0, Confidence: 0.5169)

Review excerpt: "this powerful 1995 film takes on one at the top of that list... it does it objectively and realistically, and with a sensibility and sensitivity that makes it a truly

great film by anyone's measuring stick... it is wrongly perceived that this is a film about the pros and cons of the death penalty; it is not... I rate this one 10/10."

The film has been rated a perfect 10 and is described as a ‘truly great film’. The reviewer uses the words ‘death penalty’, ‘misunderstood’, and ‘heinous crime’. The film deals with serious topics and the reviewer is not saying anything negative about the film. The model interpreted these words as the wrong label. The model’s confidence score was 0.5169. The model, in this case, recognized the positive signals, which were ‘10/10’ and ‘great film’. However, the negative sentence structures were too strong for the positive signals to overcome.

Example (error_analysis_roberta.json sarcasm part forth sentence) — RoBERTa (True: 1, Predicted: 0, Confidence: 0.9904)

Review excerpt: "We don't usually warm up to a series until they are almost at the end of their production life... By the time we got hooked, there were only a couple more season's left before the series was canceled... My only criticism is the writing wasn't always consistent... But I loved the casting and the characters are all quite believable and realistic... This is a great series and I was sorry when it cancelled."

The reviewer clearly loves the show. The reviewer writes "This is a great series" in the end of the review. But the text also contains many negation words like "didn't", "not", "canceled", and "wasn't always consistent". RoBERTa focused on those words. It predicted negative with a very high confidence of 0.9904. It completely missed the positive conclusion at the end. This is a typical negation error. The model picked up on the negation markers but did not understand the overall sentiment of the review.

7. Screenshots of Output

This section will include screenshots of evaluation system outputs.

```
=== SentiWordNet Baseline ===
              precision    recall  f1-score   support

    Negative       0.76      0.32      0.45     12500
    Positive       0.57      0.90      0.70     12500

    accuracy                           0.61     25000
   macro avg       0.66      0.61      0.58     25000
weighted avg       0.66      0.61      0.58     25000

Confusion Matrix:
[[ 4062  8438]
 [ 1295 11205]]
```

Figure 3: Classification report and confusion matrix for the SentiWordNet baseline on the IMDb test set.

```
=== RoBERTa (zero-shot) ===
              precision    recall  f1-score   support

    Negative       0.76      0.94      0.84     12500
    Positive       0.92      0.70      0.80     12500

    accuracy                           0.82     25000
   macro avg       0.84      0.82      0.82     25000
weighted avg       0.84      0.82      0.82     25000

Confusion Matrix:
[[11780   720]
 [ 3722  8778]]
```

Figure 4: Classification report and confusion matrix for the RoBERTa zero-shot model on the IMDb test set.

```
=== DistilBERT-SST2 (zero-shot) ===
              precision    recall  f1-score   support

    Negative       0.87      0.92      0.90     12500
    Positive       0.92      0.86      0.89     12500

    accuracy                           0.89     25000
   macro avg       0.89      0.89      0.89     25000
weighted avg       0.89      0.89      0.89     25000

Confusion Matrix:
[[11498  1002]
 [ 1695 10805]]
```

Figure 5: Classification report and confusion matrix for the DistilBERT-SST2 zero-shot model on the IMDb test set.

```
=== DistilBERT (fine-tuned) ===
              precision    recall  f1-score   support

    Negative       0.80      0.68      0.73     12500
    Positive       0.72      0.83      0.77     12500

    accuracy                           0.75     25000
   macro avg       0.76      0.75      0.75     25000
weighted avg       0.76      0.75      0.75     25000

Confusion Matrix:
[[ 8475  4025]
 [ 2167 10333]]
```

Figure 6: Classification report and confusion matrix for the DistilBERT fine-tuned model on the IMDb test set.

```
=== RoBERTa (fine-tuned) ===
              precision    recall  f1-score   support

    Negative       0.86      0.66      0.74     12500
    Positive       0.72      0.89      0.80     12500

    accuracy                           0.77     25000
   macro avg       0.79      0.77      0.77     25000
weighted avg       0.79      0.77      0.77     25000

Confusion Matrix:
[[ 8205  4295]
 [ 1344 11156]]
```

Figure 7: Classification report and confusion matrix for the RoBERTa fine-tuned model on the IMDb test set.

The Gradio web interface (demo/app.py) showing a sample prediction with sentiment label and confidence score, and the model selector dropdown. Show in the Figure 8.

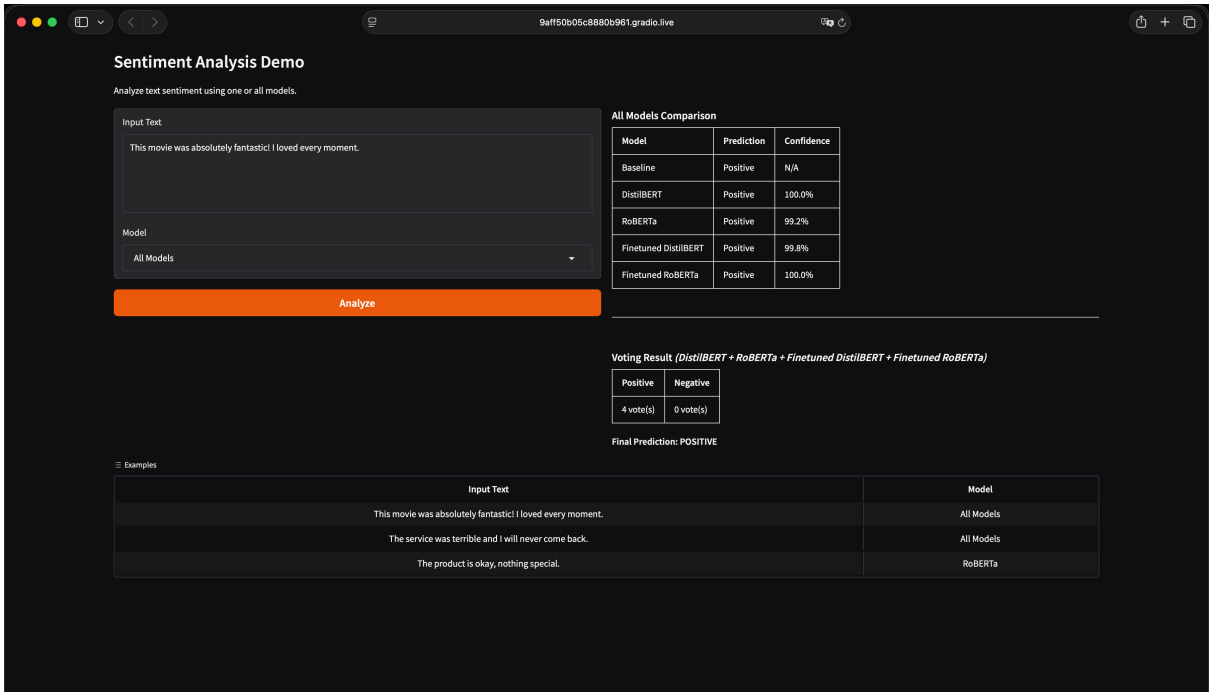

Figure 8: Gradio demo

Terminal output from demo/predict.py showing a sample text classified with a chosen model via the –text and –model flags. Showing in the Figure 9.

```
(6713_ass) zekai_gao@DESKTOP-8254V9I:~/Programs/comp6713-sentiment-analysis$ python demo/predict.py --text
"The service was terrible and I will never come back." --model finetuned-roberta
____________________________________________________
Text: The service was terrible and I will never come back.
Predicted Label: 0
Label Text: negative
Confidence: 0.9981
```

Figure 9: CLI inference

8. Overall Findings

Across all five models, DistilBERT-SST2 (zero-shot) achieved the best results at 89.2% accuracy and an F1 of 0.889, while the SentiWordNet Baseline performed worst at 61.1% accuracy. The central finding is that fine-tuning on GoEmotions did not help both fine-tuned models scored well below the zero-shot DistilBERT-SST2, confirming that domain alignment between training and test data matters more than the fine-tuning process itself.

The qualitative analysis of 60 misclassified examples (30 per fine-tuned model) found that ambiguous cases were the most common error type (32% of all errors). Many of the models' mistakes come from reviews that are genuinely hard to classify, including 'so bad it's good' reviews, mixed praise-and-criticism write-ups, and cases where the ground truth label itself seems debatable. Domain mismatch (18%) and Other multi-referent errors (22%) were the next most significant categories, where the models were misled by non-standard review styles or by positive language directed at different films. Sarcasm (17%) and negation (12%) were less frequent but often produced high-confidence wrong predictions.

One notable difference between the two fine-tuned models is that DistilBERT had more ambiguous and sarcasm errors (11 and 6 out of 30), while RoBERTa had more domain mismatch and Other errors (7 and 7 out of 30). Both models struggled in cross-domain transfer, but for slightly different reasons.

9. Limitations

There are several limitations in this evaluation. The cross-domain setup makes it difficult to fairly compare models that have different levels of domain alignment with the test set. The qualitative analysis is based on 60 manually inspected samples, which may not capture the full range of error patterns across 25,000 test examples. Binary sentiment labels also oversimplify the task, as the large ambiguous category shows, many reviews sit genuinely between positive and negative, and the binary ground truth label is itself sometimes questionable.